OIL COOLER REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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Warning

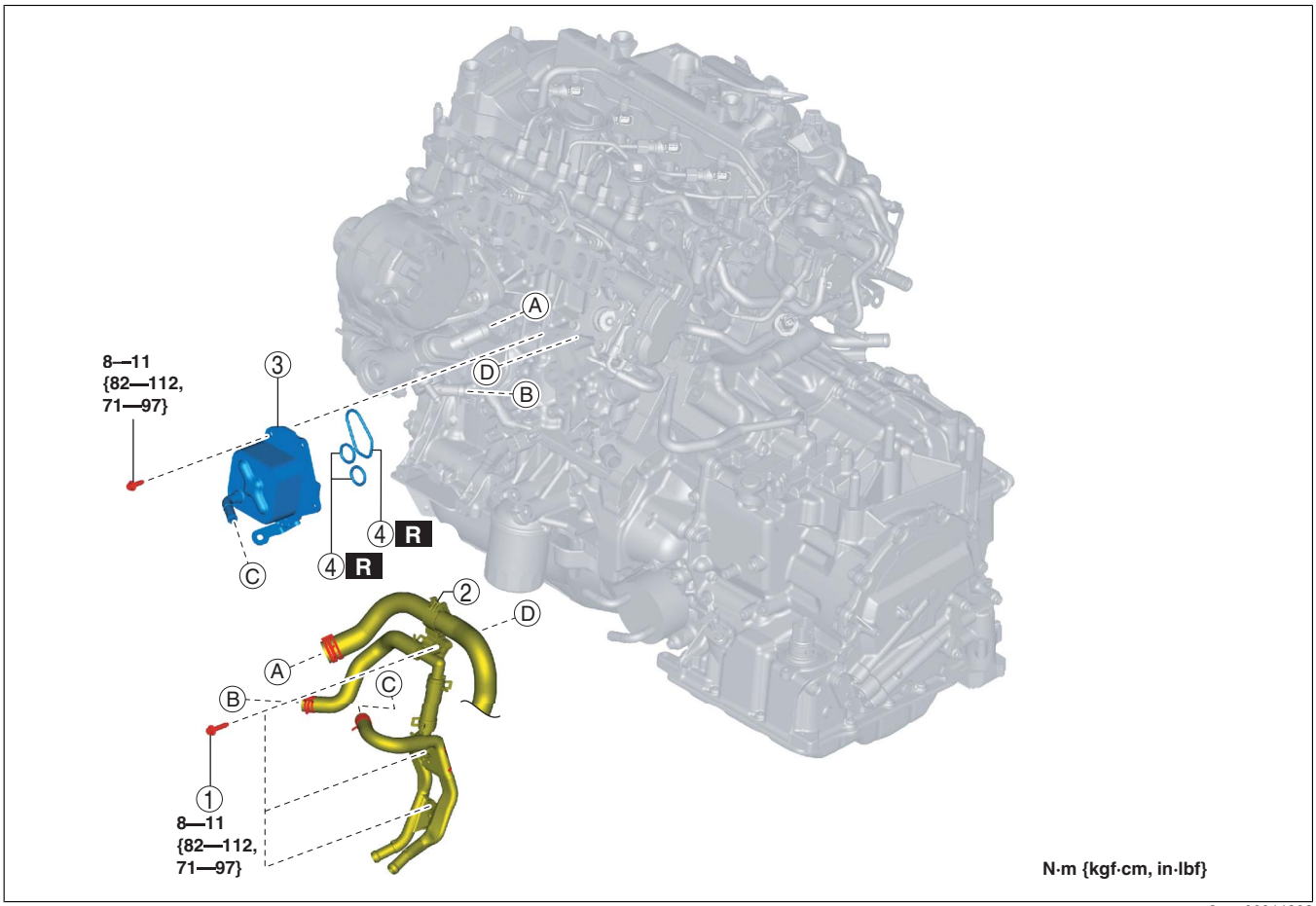
- **Hot engines and engine oil can cause severe burns. Turn off the engine and wait until it and the engine oil have cooled.**
- **A vehicle that is lifted but not securely supported on safety stands is dangerous. It can slip or fall, causing death or serious injury. Never work around or under a lifted vehicle if it is not securely supported on safety stands.**
- **Continuous exposure to USED engine oil has caused skin cancer in laboratory mice. Protect your skin by washing with soap and water immediately after working with engine oil.**

Note

- Remove the following parts before removing/installing the oil cooler.
  - — ATX: Intake manifold component, electric water pump, starter
  - — MTX: Intake manifold component

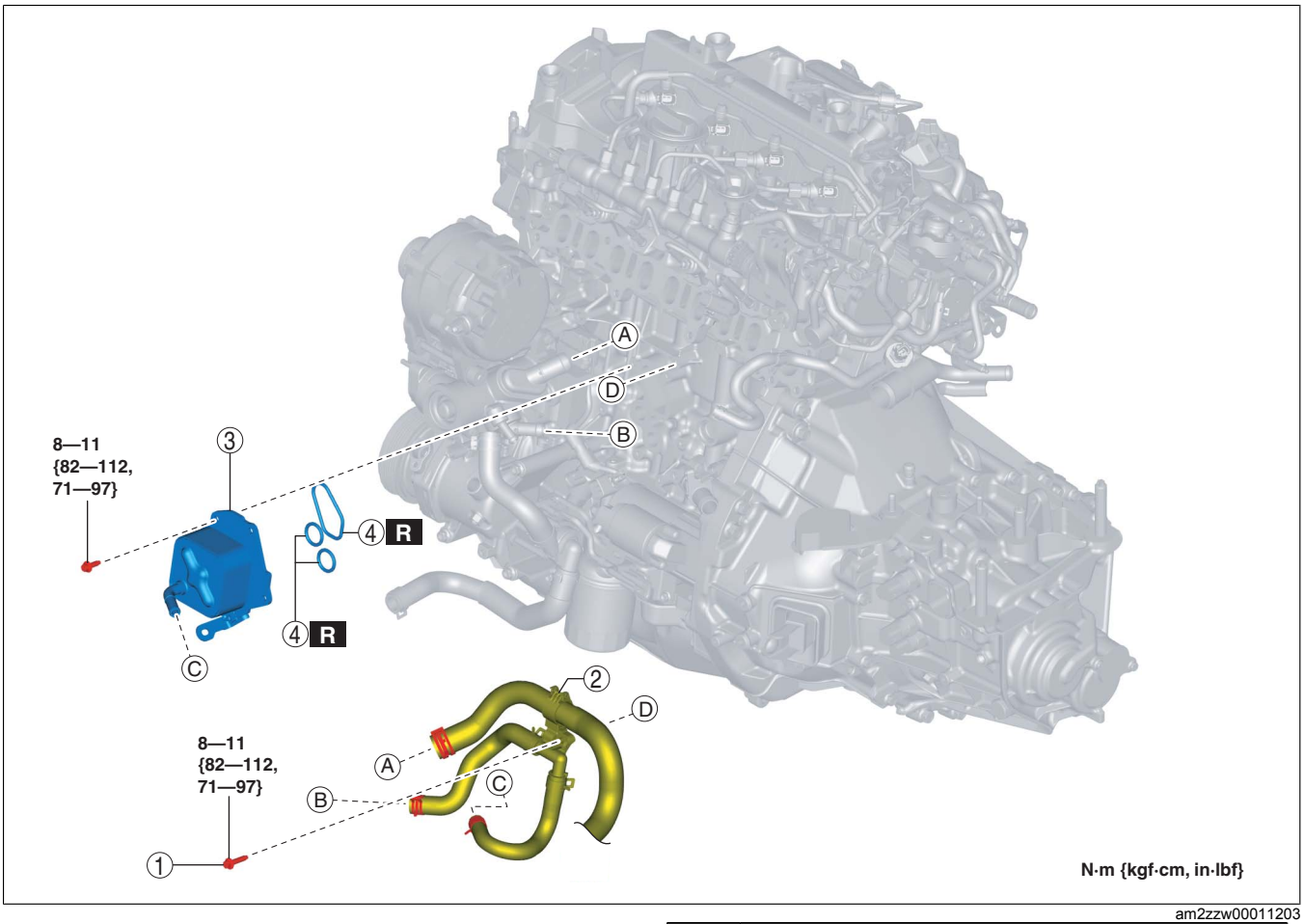
1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the front under cover No.1 and No.2. (See FRONT UNDER COVER No.1 REMOVAL/ INSTALLATION.)(See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Drain the engine coolant and the water-cooled charge air cooler coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
5. Remove the intake manifold component. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Remove the electric water pump.(ATX) (See ELECTRIC WATER PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7. Remove the starter.(ATX) (See STARTER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8. Remove in the order indicated in the table.
9. Wipe engine oil or coolant off completely from the following parts.
   - Oil cooler installation surface on the cylinder block side
   - O-ring installation groove and installation surface on the oil cooler
10. Install in the reverse order of removal.
11. Refill the engine coolant and water-cooled charge air cooler coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
12. Start the engine and confirm that there is no oil leakage.
    - If there is oil leakage, repair or replace the applicable part.
13. Inspect the oil level. (See ENGINE OIL LEVEL INSPECTION [SKYACTIV-D 1.5].)
14. Inspect for engine coolant leakage and water-cooled charge air cooler coolant leakage. (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)

ATX



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MTX



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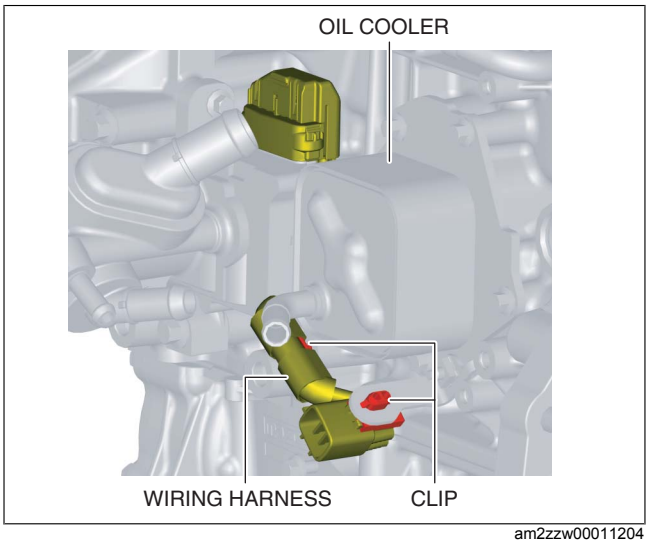
1	Water pipe installation bolts
2	Water hose and pipe component (See Water Hose and Pipe Component Removal Note.)
3	Oil cooler (See Oil Cooler Removal Note.) (See Oil Cooler Installation Note.)
4	O-ring (See O-ring Installation Note.)

Water Hose and Pipe Component Removal Note

1. Before disconnecting the hose, place an appropriate container under the engine to collect the engine coolant.
2. Disconnect the three water hoses (one on oil cooler part and two on coolant control valve part).
3. Set the water hose and pipe component aside so that it does not interfere with the servicing.

Oil Cooler Removal Note

1. Detach the clips shown in the figure and set the wiring harness aside.
2. Cover the area around the oil cooler with a rag because engine oil and engine coolant leak from the oil cooler installation area.
3. Remove the oil cooler.



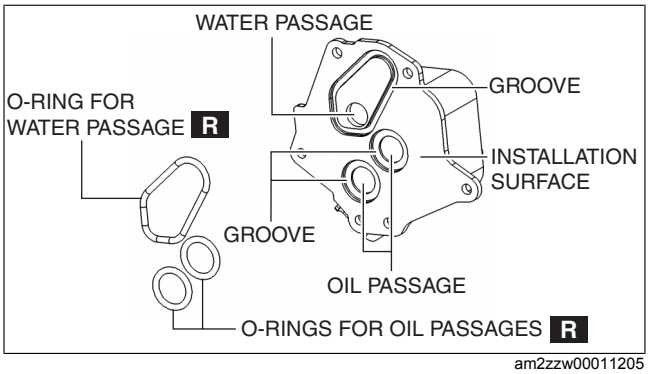
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O-ring Installation Note

1. Install new O-rings to the oil cooler.

Caution

- **Do not apply engine oil, ATF, or engine coolant to the O-ring installation groove and installation surface on the oil cooler before installing the O-rings to the oil cooler.Otherwise the O-rings could swell and cause a seal malfunction. In addition, do not apply engine coolant to the O-rings for the oil passages, and do not apply engine oil to the O-ring for the coolant passage.**

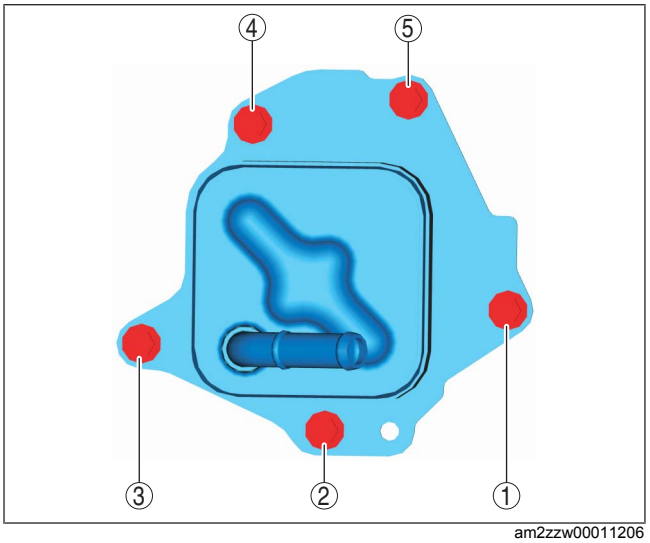


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Oil Cooler Installation Note

1. Tighten the bolts in the order shown in the figure.

Tightening torque
8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}



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